

WASP-35b

R-1177

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_240209 · created 2026-07-31T03:56:14+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460349.7103 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.048
Transit depth (Rp/Rs)^2	2.7 +/- 1.57 [%]
Orbital Inclination (inc)	88.41 +/- 2.03
Airmass coefficient 1 (a1)	1.03 +/- 0.59
Airmass coefficient 2 (a2)	-0.14 +/- 0.15
Scatter in the residuals of the lightcurve fit is	50.94 %
Best Comparison Star	None
Optimal Aperture	4.49
Optimal Annulus	5.28
Transit Duration (day)	0.133 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.048 vs 0.1241 (deviation 0.83σ)
Duration fit vs published	0.133 d vs 0.1317 d (deviation 0.1σ)
Mid-time O–C	-0.1 min
Depth SNR	3.4
Residual scatter	50.94 %

Light curve

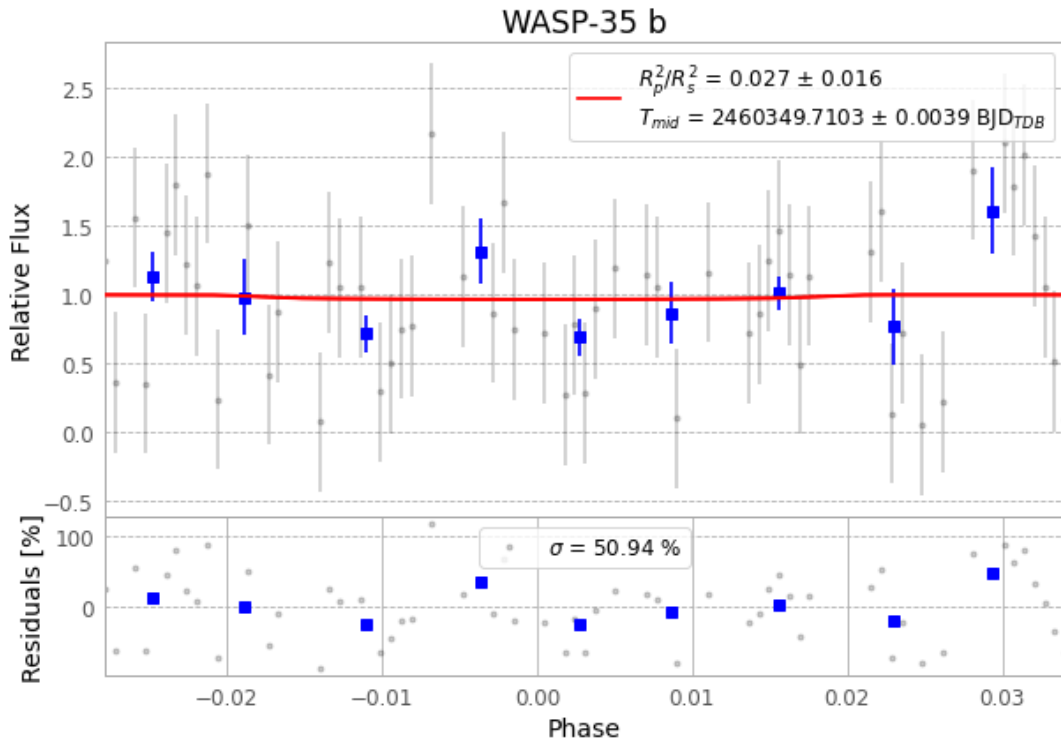


Figure 1 — FinalLightCurve_WASP-35 b_2024-02-08.png (SHA-256 2955b6c824aa7a80...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [586, 361], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8551d3c58b538568...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

95 FITS frames (62 MB), WASP-35240209025115.FITS → WASP-35240209073316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-240209.log (c1ead8679801...), FinalLightCurve_WASP-35 b_2024-02-08.png (2955b6c824aa...), FinalLightCurve_WASP-35 b_2024-02-08.csv (83abe4f9d1d1...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:56Z.